Article-Level Novelty Scoring for News Sentiment Signals: Framework and Preliminary Evidence

Richard Niemann Pjotrs Baskakovs
Yukka AG

September 10, 2026

Internal working note — not for publication

Abstract

[plan] Write last. Five moves, in this order.
(1) *The question.* A production sentiment signal counts every article about a company as one independent observation. Most of what a news feed carries is repetition, so the signal measures information arrival and information repetition jointly. Can scoring each article for novelty make the signal better, and how?
(2) *Three ways in.* A novelty score can enter a signal as a filter, as a weight, or as a selector for originating reports. They are not variants of one idea — they differ in what they discard, in how they interact with sample size, and in what they require to be built.
(3) *What we measured.* 45 settings of the cheapest score, over 518 trading days, through an unchanged production book, in weight and filter mode.
(4) *What we found.* State the results that survive the sample, and state what *cannot* be settled at this length — and therefore which endpoint the next phase should be judged on.
(5) *What is next.* A ten-year backfill on a chronologically consistent embedding model, the originator stage, and the comparative protocol.
Target 220–260 words. Do not lead with the dispersion correction; it is a measurement finding inside §2, not the paper's claim.

[plan] Reading guide for a future maintainer — unnumbered paragraph right after the abstract, half a page. Which notebook produced which figure; which sections are settled framework (§§2–3); which are evidence (§§6–8); which are unimplemented (the originator stage of Appendix C, the lag test of Appendix E, and all of §10). Someone returning after six months should know in ninety seconds what is real and what is plan.

1 Introduction

1.1 Forms and volumes of reproduction

Most of the information a news feed carries about a company has been carried before. In the corpus used in this paper, 38.9 million articles on 229 companies over two years, 57.8% of

articles repeat a headline already published about that company on the same day, and 70.2% of articles are exact or near-exact reprints of previous articles. Both figures are lower bounds, since neither detects a rewritten headline and neither observes body text. A sentiment pipeline that treats each article as an independent observation therefore counts a single event many times over.

Republication is also the mechanism by which a story reaches the market, and it carries measurable price impact. Peress (2014) found that suspending newspaper publication reduced trading volume and return dispersion, and that newspapers largely propagate the previous day's news. The article reporting an event first therefore precedes the coverage that prices it. A signal restricted to originating reports should consequently lead one built on the undifferentiated stream, with the available lead bounded by the time a story takes to propagate across outlets. Identifying that interval is the return this paper seeks, and it is reachable only once originating reports can be separated from their echo.

Separation requires a measure of degree. Verbatim reprints, translations, abridgements, aggregations of several earlier pieces and separate reporting of one event occupy a continuum, and the economically consequential case lies between its extremes. Fedyk and Hodson (2023) reported that prices respond more strongly to *recombinations* of previously published material than to direct reprints. A recombination duplicates no single predecessor, so a binary duplicate flag cannot represent it. The quantity to be measured is the novelty of each article relative to what has already been published about the same company.

1.2 The economics of coverage

The prevalence of republication follows from the cost structure of the industry producing it. Advertising revenue migrated away from news, collapsing publishers' margins while the cost of original reporting remained unchanged, and digital distribution simultaneously reduced the marginal cost of republication to a negligible level. Pickard (2020) documented these conditions for the United States. Print advertising revenue was lost at approximately ten times the rate at which digital revenue was gained, the digital-native outlets expected to replace legacy newsrooms deflated after 2017, and ownership concentration proceeded behind an appearance of outlet diversity. Underlying both developments is the non-excludability of reported facts. An outlet establishing a fact cannot prevent others from restating it, so the cost of originating is borne privately while the benefit accrues generally. The private return to originating is correspondingly low, and the equilibrium tilts toward republication.

What changes under this pressure is behaviour within outlets. A newsroom that loses reporting capacity but must still publish daily fills the gap with wire copy and rewriting, so the count of active sources holds or grows while the number of independent observations per event declines. The source count of a feed is therefore not an observer count. Origination concentrates, and Horne and Adalı (2018) found republication communities in the United States organised around a small number of agencies. That concentration opens the possibility of selecting originating reports, described further in §1.3.

The share of republished content is highest at regional outlets. A regional outlet whose reporting capacity has fallen maintains its publication schedule by taking agency copy, and that copy is predominantly national and international. Vogler and Häuptli (2018) measured declared agency copy across Swiss print and online outlets in 2016 and 2017, and the highest shares were regional, at 37 to 41% of items at three regional newspapers and 59 to 63% at two regional news sites, against 13 to 15% from the national agency at two elite titles. Only 15% of the agency items they classified were local or regional in subject. Boumans et al. (2018) traced

a year of Dutch agency output into the national titles and attributed up to 75% of online articles to the agency, a large share of it reproduced with little or no editing.

Origination is therefore a property of an outlet within a subject area and not of the outlet alone. On a widely covered event, a small set of sources would hold nearly all the originating reports. Confining an outlet's influence to the companies it originates about would bring the articles counted for a company closer to the number of independent reports about it.

The driver being cost structure, the pattern has a datable onset and can be expected to outlast it. In the United States the conditions Pickard (2020) documented were in place by the mid-2000s and deepened after the financial crisis of 2008. Measurements of text reuse in European news, from Nicholls (2019) onward, find the pattern already established there. The two years of data used in this paper open in 2024 and fall well inside that period, as does the longer history proposed in §10.

1.3 Consequences and contributions

An event's weight in a signal built on every article is therefore proportional to how widely it was syndicated, and not to the information it carried.

Weighting by syndication is costly because syndicated coverage is already priced. Tetlock (2011) measured the staleness of a story as its textual similarity to the previous ten stories about the same company, on US equities, and found that returns responded less to stale news while returns on stale-news days reversed over the following week. Using all articles therefore concentrates a signal's weight on the coverage whose news is oldest, where the measured response is smallest and where part of it is later reversed.

The mismeasurement enters when articles are counted, so a correction has to attach to the article and precede aggregation.

This paper contributes the following.

First, the paper states the problem and proposes a family of candidate measures of novelty. Each member is placed against its appearances in the literature. The two implemented candidates are given in full (§2) and the remainder are deferred (Appendix D).

Second, the paper describes the three ways such a measure can enter a signal (§3), and the protocol by which the adjusted signals are benchmarked against the unadjusted one (§10.5). The third way requires a two-stage originator criterion combining novelty with echo confirmation, which is new against the existing literature (§4.3, Appendix C).

Lastly, the paper reports pilot evidence on the filter and weight routes, with the scope of each result stated (§§7–8).

2 The novelty score

The score is determined by four choices, a map ϕ from an article to the text that is compared, a map ψ from that text to a vector together with a similarity on those vectors, a reference set $\mathscr{R}_c(d)$ of prior articles, and a function f from the similarities between d and the members of $\mathscr{R}_c(d)$ to a number in $[0,1]$. §2.1 states the problem and §§2.2–2.5 make the four choices in that order. Concrete values for the parameters introduced are fixed in §6.

2.1 Problem statement

The quantity to be measured is the novelty of an article with respect to what had already been published about a given company (§1.3).

Let $\mathscr{D}$ denote the corpus of articles. For an article $d \in \mathscr{D}$, let $t(d)$ be its publication timestamp, $o(d)$ its outlet and $C(d)$ the set of companies it mentions. Let ϕ map an article to the text used for comparison and let $x_d \in \mathbb{R}^m$ be the embedding of $\phi(d)$. Define the similarity of two articles to be the cosine similarity of their embeddings,

$$s(d,d') \;=\; \frac{x_d^\top x_{d'}}{\|x_d\|\,\|x_{d'}\|}.$$

Embeddings are normalised to unit length throughout, so $x_d \in \mathbb{S}^{m-1}$ and $s(d,d') = x_d^\top x_{d'}$.

Novelty is a property of an article paired with one of the companies it mentions, because the same article can be a first report about one company and a restatement about another. The reference set of d with respect to a company $c \in C(d)$ is a subset $\mathscr{R}_c(d) \subseteq \mathscr{D}$.

The object sought is a novelty function

$$\nu_c(d) \;=\; f\big(x_d, \{x_{d'} : d' \in \mathscr{R}_c(d)\}\big) \;\in\; [0,1],$$

small when d is a near-duplicate or close paraphrase of prior coverage of c, and large when d introduces content unlike that coverage. It should be invariant under translation into another language, under changes in length that leave the content alone, and under the addition of text common to every article from a given outlet.

Soboroff and Harman (2005) reported that assessors in the TREC Novelty track agreed poorly on which sentences were novel, and that systems barely improved on a baseline returning every relevant sentence. The quantity being estimated is therefore imprecise independently of how it is computed, which is why the requirements above are comparative and fix no value of $\nu_c(d)$ for any article.

2.2 Article representation

The map ϕ reduces an article to the text that is compared. In financial reporting the event is concentrated in the opening sentences, while company background and standing disclaimers recur across articles about the same company whatever those articles report. A map admitting that material raises the measured similarity of two articles for reasons unconnected to whether they carry the same news.

ϕ_{head} takes the headline and lede, the first two or three sentences of d. ϕ_{comp} takes the sentences of d mentioning a company in $C(d)$, each with one sentence of context on either side. ϕ_{full} takes the body of d, with navigation, advertising and standing legal text removed.

Cost is not the only reason to prefer the cheapest of the three. Kuntur et al. (2026) traced editorial reuse across news agencies and found the lede the most original part of an article, with reuse concentrated in the middle and the end. Wu et al. (2025), on academic papers, found that a subset of informative sections predicted novelty better than the whole text did. A map restricted to the lede therefore drops the part of an article most likely to have been reused.

2.3 Embeddings and similarity

Let ψ denote the embedding model, so that $x_d = \psi(\phi(d))$. Its output is normalised to unit length, under which

$$\|x_d - x_{d'}\|_2^2 \;=\; \|x_d\|^2 + \|x_{d'}\|^2 - 2x_d^\top x_{d'} \;=\; 2 - 2\,s(d,d').$$

Euclidean distance is a strictly decreasing function of cosine similarity. Both therefore order the members of $\mathscr{R}_c(d)$ identically and pick out the same n nearest neighbours. The scores built from those neighbours are not interchangeable. A mean over the n largest similarities is not determined by the corresponding mean over the n smallest distances.

The vectors have to come from a model that embeds at sentence or passage level, that is trained so the cosine between its outputs is meaningful, and that places a text and its translation close together without a translation step. Sentence-transformer models supply the first two properties (Reimers and Gurevych, 2019), and multilingual versions distilled from a monolingual teacher supply the third (Reimers and Gurevych, 2020). The model used here is named in §6.

These models are trained so that the cosine between their outputs reproduces human ratings of similarity in meaning. Similarity in meaning is not sameness of event.

Under an isotropic distribution the average cosine between randomly sampled representations would be zero. On word representations, Ethayarajh (2019) found no non-input layer of BERT, ELMo or GPT-2 isotropic, with that average at roughly 0.6 through the middle layers of GPT-2 and close to 1 at the last. Gao et al. (2019) attribute the geometry to the training objective, so the level at which unrelated pairs sit is determined by the model. A threshold placed on a raw score therefore carries no meaning beyond the model that produced it. Standardising or whitening the embeddings would correct the scale without making the target precise (§2.1).

The problem is not specific to embeddings. Montague and Aslam (2001) state the general case, that raw scores from different sources are not on a comparable scale and must be normalised before being combined or thresholded.

The score is therefore calibrated before any threshold is applied, replacing an absolute boundary with a relative one. Write $\widehat{G}_c$ for the empirical distribution of raw novelty over d's reference set,

$$\widehat{G}_c(u) \;=\; \frac{1}{|\mathscr{R}_c(d)|} \sum_{d' \in \mathscr{R}_c(d)} \mathbf{1}\{\nu_c(d') \le u\},$$

and set $\tilde{\nu}_c(d) = \widehat{G}_c\big(\nu_c(d)\big)$, the share of that reference set whose own novelty does not exceed d's. The calibrated score takes values in multiples of $1/|\mathscr{R}_c(d)|$, and $\tilde{\nu}_c(d)$ is invariant under any strictly increasing transformation of ν_c.

The form of the transformation has precedent as well. Manmatha et al. (2001) map a raw score to its position within a fitted distribution, which is the mechanism used here. Webber et al. (2008) report the same reasoning in the domain of system effectiveness scoring. Standardising each score against its observed distribution made them comparable across collections.

2.4 Reference set

The reference set is the set of prior articles an article is compared against, and every novelty score in this paper is computed relative to it. Write

$$\mathscr{D}_t \;=\; \{\, d' \in \mathscr{D} \;:\; t(d') < t \,\}$$

for the corpus published before time t, and call $d' \in \mathscr{D}_{t(d)}$ an antecedent of d with respect to c if part of what d reports about c was already reported in d'. The reference set should satisfy

1. $\mathscr{R}_c(d) \;\subseteq\; \mathscr{D}_{t(d)}$,
2. $\mathscr{R}_c(d) \;\subseteq\; \{\, d' \in \mathscr{D} \;:\; c \in C(d') \,\}$.

Remark. (1) ensures that $\nu_c(d)$ is computable at $t(d)$, since no article published later enters the set. (2) ensures that every similarity entering $\nu_c(d)$ is between two accounts of c.

Within (1) and (2) the set could still have gaps in time. Assume that an earlier article is the more likely to be an antecedent of d the more recently it appeared. Leskovec et al. (2009) tracked quoted phrases through ninety million US news and blog articles and found phrase volume decaying exponentially away from its peak with a half-life of roughly eight hours. In their generative model a story's probability of being covered declines with its age, and the model does not reproduce the observed dynamics without that decline. Under the assumption the k most recent articles are the k most probable antecedents for every k, so a set with a gap is improved by closing it and dropping its oldest members instead. The set is therefore contiguous, and what remains open is how far back it reaches.

If every antecedent of d predates the earliest member of the set, d is compared only with articles that carry none of what it reports about c, so $\nu_c(d)$ is high and a restatement is scored as novel. A set that reaches past the period at which corporate events recur admits the previous instance of a recurring event. A quarterly report is more similar to the previous quarter's report than to any other article about the company, and any function meeting the requirements of §2.1 must then score it as a near-duplicate, so a report that repeats nothing is scored as a restatement. Taking novelty as the positive class, the first error is a false positive and the second a false negative.

The reach is set by a lookback duration w, so that for an article d and a company $c \in C(d)$

$$\mathscr{R}_c(d;w) \;=\; \big\{\, d' \in \mathscr{D} \;:\; c \in C(d'),\;\; t(d) - w \le t(d') < t(d) \,\big\}.$$

Write $n_c(d) = |\mathscr{R}_c(d;w)|$ for its size, and $s_{c,(1)}(d) \ge s_{c,(2)}(d) \ge \cdots \ge s_{c,(n_c(d))}(d)$ for the similarities between d and its members, in decreasing order.

2.5 Candidate functions

Define, for a neighbour count $k \in \mathbb{N}$,

$$\nu_c^{(k)}(d) \;=\; 1 - \frac{1}{\min\{k,\, n_c(d)\}} \sum_{j=1}^{\min\{k,\, n_c(d)\}} s_{c,(j)}(d),$$

and set $\nu_c^{(k)}(d) = 1$ when $\mathscr{R}_c(d;w)$ is empty, since no earlier account of c is then within reach.

At $k = 1$ the function is the nearest-neighbour score $\nu_c^{(1)}(d) = 1 - s_{c,(1)}(d)$. It compares d with the one member of the reference set most similar to it, and no other member enters. Allan et al. (1998) use the same comparison and declare a document a new event if its similarity to every earlier document is below a threshold. The score measures whether any single earlier article already carried what d reports about c.

At $k > 1$ the function is the top-k mean. It compares d with its k most similar members and measures how much of what d reports is carried by those k together. Vuurens and de Vries (2016) found that the mean over the three nearest neighbours detected first stories better than the distance to the single nearest neighbour. Both functions are calibrated as in §2.3 before any threshold is applied.

The top-k mean admits a second reading. Let $\hat{x}_c^{(k)}(d)$ be the mean direction of the embeddings of the k members of $\mathscr{R}_c(d;w)$ most similar to d. By linearity of the inner product, $\nu_c^{(k)}(d)$ equals the centroid score $1 - x_d^{\top}\hat{x}_c^{(k)}(d)$ up to a multiplicative factor, and at $k = n_c(d)$ the centroid is that of the whole reference set (Appendix B). Four further candidates are neither implemented nor evaluated and are described in Appendix D.

3 Aggregation schemes

A novelty score is a property of one article. A signal is a property of a company and a day, so the score enters the signal only through the step that combines articles, and there are three places it can enter. Let $\mathrm{tone}(d) \in \{-1, 0, +1\}$ be the tonality assigned to article d, and for a company c and a trading day t define

$$A_{c,t} \;=\; \big\{\, d \in \mathscr{D} \;:\; c \in C(d),\;\; t(d) \in t \,\big\},$$

the articles about c published on t, where t is taken as the interval from the previous trading day's close to its own. Every member has $c \in C(d)$, so $\tilde{\nu}_c(d)$ is defined for each. Write

$$P_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=+1}} 1, \qquad N_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=-1}} 1$$

for the numbers of positive and negative articles. The signal in place is

$$S_{c,t} \;=\; \frac{P_{c,t}+1}{P_{c,t}+N_{c,t}+2} \;\in\; (0,1),$$

the positive share with a Laplace prior, in which every signed article counts once and neutral articles do not enter.

3.1 Filtering

For a threshold $\theta \in [0,1]$ define

$$P^{\theta}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=+1 \\ \tilde{\nu}_c(d) \geq \theta}} 1, \qquad N^{\theta}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=-1 \\ \tilde{\nu}_c(d) \geq \theta}} 1, \qquad S^{\theta}_{c,t} \;=\; \frac{P^{\theta}_{c,t}+1}{P^{\theta}_{c,t}+N^{\theta}_{c,t}+2}.$$

Filtering removes every article with $\tilde{\nu}_c(d) < \theta$, so the tonality of a repeated article does not enter the signal at all. Ke et al. (2019) split newswire articles into fresh and stale by their cosine similarity to the company's articles over the preceding five trading days and found the return response to sentiment larger for fresh articles, and priced over more days.

3.2 Weighting

Replace each count by the novelty mass of the articles counted,

$$P^{\mathrm{w}}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=+1}} \tilde{\nu}_c(d), \qquad N^{\mathrm{w}}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \\ \mathrm{tone}(d)=-1}} \tilde{\nu}_c(d), \qquad S^{\mathrm{w}}_{c,t} \;=\; \frac{P^{\mathrm{w}}_{c,t}+1}{P^{\mathrm{w}}_{c,t}+N^{\mathrm{w}}_{c,t}+2}.$$

Every signed article enters, weighted by its novelty. The route assumes that the information an article carries increases with its novelty over the whole range and not only across one boundary. Fedyk and Hodson (2023) found that prices responded more to articles combining content from several earlier articles than to reprints of one. The response to repetition is therefore graded, and a threshold gives both kinds weight zero. $S^{\mathrm{w}}_{c,t}$ depends on $\tilde{\nu}_c(d)$ for every signed $d \in A_{c,t}$, whereas $S^{\theta}_{c,t}$ depends on it only through whether it exceeds θ. The calibration of §2.3 therefore bears on this route at every article and on the filter route only at the threshold.

3.3 Originator selection

The third route counts only articles that are first reports. Whether an article is a first report is established from two conditions, that it is novel with respect to the coverage of c that preceded it, and that it is echoed by the coverage that follows. For an article d'' with $c \in C(d'')$ let

$$\pi_c(d'') \;=\; \operatorname*{arg\,max}_{d' \in \mathscr{R}_c(d'';w)} \; s(d'',d')$$

be its most similar member of its reference set, and define $\pi_c^1 = \pi_c$ and $\pi_c^j = \pi_c \circ \pi_c^{j-1}$ for $j \geq 2$. Each application moves to a strictly earlier article, so the sequence $d'', \pi_c(d''), \pi_c^2(d''), \ldots$ is finite. For a confirmation horizon $\tau > 0$ define the echo set of d to be

$$\mathscr{E}_c(d;\tau) \;=\; \big\{\, d'' \in \mathscr{D} \;:\; c \in C(d''),\;\; t(d) < t(d'') \leq t(d) + \tau,\;\; \pi_c^j(d'') = d \text{ for some } j \geq 1 \,\big\},$$

the later articles about c from which d is reached by repeatedly moving to the most similar prior article. An article whose nearest predecessor is itself an echo of d is thereby an echo of d. For a threshold θ and a minimum echo count $q \in \mathbb{N}$, d is an originating report about c if

$$\tilde{v}_c(d) \;\geq\; \theta \qquad \text{and} \qquad |\mathscr{E}_c(d;\tau)| \;\geq\; q.$$

The first condition alone does not suffice, because translation artefacts, off-topic mentions and low-quality reproductions all score as novel and none is a first report. Write

$$\mathscr{F}_c \;=\; \big\{\, d \in \mathscr{D} \;:\; c \in C(d),\;\; \tilde{v}_c(d) \geq \theta,\;\; |\mathscr{E}_c(d;\tau)| \geq q \,\big\}$$

for the set of originating reports about c, and define

$$P^{\mathrm{s}}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \cap \mathscr{F}_c \\ \mathrm{tone}(d) = +1}} 1, \qquad N^{\mathrm{s}}_{c,t} \;=\; \sum_{\substack{d \in A_{c,t} \cap \mathscr{F}_c \\ \mathrm{tone}(d) = -1}} 1, \qquad S^{\mathrm{s}}_{c,t} \;=\; \frac{P^{\mathrm{s}}_{c,t} + 1}{P^{\mathrm{s}}_{c,t} + N^{\mathrm{s}}_{c,t} + 2}.$$

The set $\mathscr{E}_c(d;\tau)$ is known only at $t(d) + \tau$, so a live signal cannot evaluate the second condition. It uses instead the rate at which an outlet has originated. For an outlet o define

$$\gamma_{c,o} \;=\; \frac{|\{\, d \in \mathscr{F}_c : o(d) = o \,\}|}{|\{\, d \in \mathscr{D} : o(d) = o,\; c \in C(d) \,\}|},$$

the share of o's articles about c that were first reports, and

$$\gamma_o \;=\; \frac{\sum_c |\{\, d \in \mathscr{F}_c : o(d) = o \,\}|}{\sum_c |\{\, d \in \mathscr{D} : o(d) = o,\; c \in C(d) \,\}|},$$

the same share pooled over all companies o covers. §1.2 found origination to be a property of an outlet within a subject area, so the rate is taken per outlet and company, and shrunk toward γ_o where the pair has too few articles. The live signal then weights each article by $\gamma_{c,o(d)}$ in place of $\tilde{v}_c(d)$ in §3.2. Nothing in this route is implemented. The outlet-level quantities it yields beyond $\gamma_{c,o}$ and the ways it can fail are given in Appendix C.

4 Related work

Prior work is sorted by the route of §3 it uses. The filter route has three precedents, the weight route none that weights by novelty, and the originator route none.

4.1 Filtering

Tetlock (2011) defined the staleness of a story as its textual similarity to the previous ten stories about the same company, which is the reference set of §2.4 with its reach set by an article count in place of w, and found on US equities that returns responded less to stale stories and that returns on stale-story days reversed over the following week. Ke et al. (2019) split newswire articles into fresh and stale by cosine similarity to the company's articles over the preceding five trading days at a fixed cutoff and found the return response to sentiment larger for fresh articles, and priced over more days. The split is the filter of §3.1 with an uncalibrated threshold. RavenPack Data Science Team (2014) filtered a sentiment indicator on the time since a similar event for the same company, at a gap of ninety days, and reported the long/short Sharpe ratio on the Russell 2000 rising from 1.03 to 1.43 over 2007–2014. The note is vendor research and not peer reviewed, and its measure of similarity is recurrence of an event category and not distance between texts, so it does not detect repetition within a category. A positive result on the filter route in this paper is therefore a replication on European equities and with a calibrated score.

4.2 Weighting

No located work weights articles by a novelty score before aggregation, but several apply a continuous per-article transformation of the same kind to improve the predictive power of sentiment. Glasserman and Mamaysky (2019) measured the unusualness of an article as the negative log-likelihood of its word sequences under a language model fitted to the preceding year of articles, and used the product of unusualness and sentiment as the predictor. Unusual negative news about large financial firms forecast higher volatility and unusual positive news lower volatility, over horizons of months. Mizuno et al. (2017) constructed a novelty index and a topicality index, the latter the number of agencies covering a story at the same time and so a count of the kind $\mathscr{E}_c(d;\tau)$ measures, and related both to trading volume and volatility on the same day without testing either against subsequent returns. Didisheim et al. (2026) removed from each article the content predictable from company characteristics and found that the residual more than doubled the monthly return predictability of news, with the effect persisting over eighteen months. The residual is taken against characteristics, whereas $\nu_c(d)$ is taken against prior news about the same company, and whether the second is subsumed by the first is a question the protocol of §10.5 answers. In machine learning, Wang and Ma (2024) found that averaging article sentiments over a day homogenises the aggregate and proposed learned per-article weights. The weight route of §3.2 fixes the weight to be novelty in place of learning it.

4.3 Originator selection

No located work scores articles for novelty and then designates first reports by requiring later articles to have them as nearest prior neighbour. The components exist separately. Leskovec et al. (2009) clustered variants of quoted phrases across US news and blogs and measured the lag by which blogs followed mainstream media, which is the lead-time output of Appendix C at the level of the phrase and without a novelty score. Gomez-Rodriguez et al. (2010) inferred a diffusion network from observed adoption times alone, allowing several parents per node, where π_c assigns one. Horne and Adalı (2018) mapped verbatim republication between news producers into a who-copies-whom network centred on the Associated Press. Kuntur et al. (2026) detected sentence-level reuse between the Slovenian Press Agency and fifteen foreign agencies in seven languages and retained the earliest source by timestamp for each reused sentence, finding that one-to-one reuse made up under five per cent of cases. Nicholls (2019) sep-

arated newswire copy, press-release copy, outlet-to-outlet copying and common-source overlap in shared text between news articles, four classes that $\mathscr{E}_c(d;\tau)$ does not distinguish. In finance, Bushee et al. (2010) and Drake et al. (2014) measured the effect of press coverage on information asymmetry and mispricing around earnings, and Twedt (2016) the effect of newswire dissemination on the speed of price discovery, without attributing articles to an originating outlet.

4.4 Novelty measures

Table 1 places each candidate of §2.5 and Appendix D against the work in which the same measure, or its immediate ancestor, appears. Neither implemented candidate is new. The nearest-neighbour score is the rule of Allan et al. (1998), and Zhang et al. (2002) evaluated it against four other redundancy measures for adaptive filtering. The top-k mean follows Vuurens and de Vries (2016). Of the deferred candidates, the subspace residual is the measure of Hoffmann (2007) with a local subspace, and the asymmetric coverage measure has its asymmetry from Broder (1997) and Tsai et al. (2010) and its strongest existing implementation in Ai et al. (2024).

Candidate	Prior occurrences
Nearest neighbour ($k = 1$)	Allan et al. (1998), a document is a new event if its similarity to every prior document is below a threshold. Zhang et al. (2002), maximum similarity to any delivered document as one of five redundancy measures.
Top-k mean	Vuurens and de Vries (2016), three nearest neighbours in place of one for first-story detection.
Centroid (Appendix B)	Karkali et al. (2013), scoring against stream statistics without pairwise comparison.
Soft aggregation (Appendix D)	None located.
Subspace residual (Appendix D)	Hoffmann (2007), reconstruction error with respect to a principal subspace as the novelty measure. Mesarcik et al. (2022), reconstruction from the nearest neighbours.
Time-decayed kernel (Appendix D)	None located.
Asymmetric coverage (Appendix D)	Broder (1997), containment as the asymmetric set-overlap measure. Tsai et al. (2010), the symmetric/asymmetric split for sentence-level novelty metrics. Ai et al. (2024), atomic-unit asymmetric coverage against a historical pool with salience weighting.

Table 1: Each novelty candidate against the prior work in which the same measure, or its immediate ancestor, appears.

4.5 Counterarguments

[plan] Grouped by objection, not by paper. State these before anyone else does; they are the reason the answer might be no. Three groups.

A. Repetition is priced, so filtering it discards signal. The core objection, and it arrives at three levels. Blankespoor et al. (2020) give the mechanism: because processing costs exist, a disclosure is not truly public on release, so learning is a costly choice and repetition genuinely lowers its cost. Bushee et al. (2010) and Drake et al. (2014) give the accounting evidence: the business press works through *dissemination breadth* rather than creation of new information, and coverage reduces information asymmetry and mispricing through breadth. Peress (2014) gives the direct price evidence: newspaper strikes cut volume by 12% and return dispersion by roughly 7%, and newspapers demonstrably propagate the *previous day's* news.

The payoff of grouping them — and the reason this is not just tidier — is that the objection bears very unevenly on the three routes of §3. It is close to fatal for the filter route, which discards echo outright. It is much weaker against the weight route, which discards nothing and merely re-scales, so dissemination still enters the signal at reduced weight. And it is largely orthogonal to the select route, which is about attribution rather than exclusion. That asymmetry is an argument for prioritising the weight and select routes, and it is invisible when these papers are listed one per bullet.

B. The effect may be a documented coverage or attention effect in disguise. Hillert et al. (2014): coverage volume — largely echo volume — predicts momentum with a long-run reversal. Fang and Peress (2009): no-coverage stocks earn higher risk-adjusted returns, concentrated in small, retail-held, low-analyst names. Da et al. (2011): attention rises predict prices over two weeks and then reverse. Because novelty and coverage intensity are mechanically correlated, and coverage intensity correlates with size, any novelty result must clear a coverage-count and size control before it can be called a novelty result. This is the objection §7.4 has to answer, and as of this draft it is the leading explanation of the pilot's own pattern of results — Sharpe up, information coefficient flat, beta down — so cross-reference it there rather than treating it as hypothetical.

C. The construct may not be identifiable from what we have. §2.1 already states Soboroff and Harman (2005)'s finding and §2.3 already draws the calibration consequence, so **do not restate either here**. What this section adds is the stronger reading, that the target may be too ill-defined to estimate at all, which is a different claim from the one calibration answers. Alongside it the internal version: Khan (2026) already filters outlets by t-statistic, and if that filter implicitly selects originators then novelty adds nothing the existing quality panel does not already capture. This group is the bridge to §5.2, and the orthogonality test of §10.5 is its direct test.

Add, and it is the honest counterweight to §1.2's geographic argument. Beckers et al. (2019) measured pairwise story overlap across nine Flemish newspapers in 1983, 1993, 2003 and 2013 and *rejected* the declining-diversity hypothesis: average overlap was highest in 1983 ($\rho = .092$) and lower in 2013 ($\rho = .078$). Ownership predicted similarity where time did not, with the strongest pair reaching $\rho = .415$. Two caveats to state alongside it, because they bound what it refutes: regional sections were excluded from their sample, and the study measures topic overlap between titles rather than the reuse of text. It is evidence that convergence tracks ownership rather than a secular trend, and it is not evidence that republication is rare.

Moved out. Heston and Sinha (2017) — daily news predicts one to two days while weekly aggregated news predicts a quarter, so horizon extension can come from temporal aggregation alone — is an objection to the reporting-lag horizon claim, not to this paper's thesis. It lives with the other rival explanations in Appendix E.

5 Evaluation design

[plan] NEW SECTION, placed deliberately before any result. The paper asks whether novelty scoring can make the signal better; this says what better means, what we would accept as evidence, and what we would accept as evidence against. Roughly 1.5 pages. It also absorbs the hypotheses and the null, which paper3 had buried in its evaluation section.

5.1 Choice of endpoint

[plan] Start with the arithmetic, because it drives the choice, but state it as a formula rather than as a figure. Required sample scales as $T_{\text{req}} = T_{\text{obs}}(t_{\text{target}}/t_{\text{obs}})^2$, so an effect observed at a small fraction of one standard error implies a required history that grows with the *square* of the shortfall. Substituting the pilot's numbers (§7.7) gives a requirement on the order of centuries. That is not a finding, it is a *reductio*: the unpaired Sharpe comparison is not an instrument that can be pointed at this question, and no further slicing of the same days changes that.
[TODO: redo the arithmetic on the *paired* beta-hedged difference, whose standard error comes from the moving-block bootstrap of §7.7; pairing removes the common market component so the requirement is much smaller. Report both in §7.7, and keep this subsection free of figures.]
Hence the ranked endpoints for this project:

1. **Information coefficient and its decay shape** — uses every company-day rather than one portfolio return series, and is the only quantity directly comparable to Khan (2026). Primary.
2. **IC per article** — a variant reaching baseline IC on a fraction of the volume is a cost result as well as a signal result, and is where a filter should show up first.
3. **Originator-attribution accuracy** against the self-declared labels from explicit attribution phrases — a classification problem with tens of thousands of instances that needs no return data at all. The best-powered endpoint available, and it only exists for the select route.
4. **Alpha and Sharpe** — retained for completeness and reported with the power caveat attached, never as the headline.

5.2 Hypotheses and null

[plan] Two hypotheses and one null. The paper states hypotheses only for the routes it has evidence about — the same discipline applied to notation in §2.1. The select route has no hypothesis here because it has no implementation (Appendix C), and the deferred reporting-lag claim carries its own in Appendix E.
A short table, not prose. Columns: ID, claim, direction, where tested, what would falsify it. Both rows were specified before the pilot ran, and the table should say so — it is the only thing that distinguishes them from the post-hoc readings that appear in §8.
H1 (filter) Novelty filtering raises the information coefficient per article and concentrates predictive content at shorter horizons.
H2 (weight) Novelty weighting raises the information coefficient without reducing article volume.
The null, which is a live possibility and not a formality. **[reuse: paper3 §6.4's closing sentence.]** Novelty adds no incremental predictive content over the production signal and the existing outlet t-statistic filter. Khan (2026) already selects outlets on their t-statistic, and if that filter implicitly selects originators then novelty adds nothing the existing quality panel does not already capture — which is why the orthogonality test of §10.5 is the cheapest possible falsification of the whole project and should be run before anything expensive is built. **Commit here, in writing, to reporting the null if it holds.** That sentence is the most credible thing in the document and it costs nothing to keep. Statistical nulls belong with the metrics rather than here: mean alpha difference zero, rank correlation zero, cell beta equal to production's.

6 Data and implementation

[plan] A reproducibility record — everything needed to re-run the notebooks. 2.5 pages, prose not bullets, but complete.

6.1 Score specification

The pilot computes

$$\nu_n(d) \;=\; 1 - \frac{1}{n}\sum_{j=1}^{n} s_{(j)}(d) \tag{1}$$

over headline embeddings, with entity = company and lookback duration w.

[TODO: Confirm from the scoring code whether the pilot evaluates (1) once per article and company, as §2.1 defines it, or once per article against the union of its companies' coverage. The two agree whenever $|C(d)| = 1$ and diverge otherwise, and the share of multi-company articles in the corpus bounds how much it can matter.]

[plan] State the reach plainly: (1) is the top-k-mean candidate, and at $n = 1$ the nearest-neighbour candidate. The pilot grid is that one axis of the family and nothing else.
The representation and the model, named here because §2.2 and §2.3 deliberately do not. The representation is the headline alone — the cheapest of the three defined in §2.2, and the one the evidence there suggests is a defensible default rather than a compromise. Embeddings come from OpenAI's `text-embedding-3-small`, accessed through the API.
Three consequences worth one sentence each, because they constrain what the results can mean. It is a hosted proprietary model, so the version cannot be pinned beyond its name and reproducibility depends on the provider not changing what sits behind that name — which §9 treats as a non-stationarity risk rather than a footnote. It was not trained for news-event similarity, so cosine proximity tracks general semantic similarity rather than "reports the same event", the caveat §2.3 raises for this whole class of model. And its training data predates and overlaps the evaluation window, so it has read the news it is scoring; tolerable over two years, and the first thing to fix before any longer backfill (§10.2).
Scores were not rank-calibrated (§2.3): a global threshold in filter mode, raw novelty as the weight. Say this here rather than leaving it to be inferred — it is a first-order caveat on everything in §7.

6.2 Parameter grid and signal construction

[plan] Grid: $w \in \{6\text{h}, 24\text{h}, 3\text{d}, 7\text{d}, 14\text{d}\}$ and $n \in \{1, 3, 5, 7, 10, 20, 50, 70, 100\}$, 45 cells, built with `max_cand=0` so the window is the true lookback for every company rather than a candidate cap that bites hardest on the busiest names.
Modes: filter ($\nu > \theta$) and weight (each article contributes ν in place of a count of 1) — §3.1 and §3.2. The select route is not implemented.
Signal path, identical across arms: daily counts → `sentiment_ratio` (Laplace-smoothed positive share) → `_robust_signal` (EW-median location and scale, lagged one day, clipped at ±4.2, EW-smoothed over *span*, z-scored across the cross-section) → Signal.

6.3 Portfolio construction

[plan] `nk.build_arm` pinned to `nk.PROD_PARAMS`, parameter-for-parameter from `sentiment_momentum_top200.py`: hold each name whose robust cross-sectional z exceeds 1.15, equal-weight capped at 10% per name with the residual in cash, rebalance every 10 days, scale to a 20% vol target. Universe: point-in-time top-200 STOXX 600 by rank. Span 19.
All 46 portfolios are produced by identical code at identical settings, so no difference in outcome is attributable to the book. **But the claim stops there, and the paper must not overstate it.** The production counts are filtered server-side to outlets with $\tau_o \geq 2.5$; the novelty arms are built from the raw article dump, which is not source-filtered. Production therefore draws on a quality-selected subset of outlets and the arms on the full set of roughly 140,000, so the two differ in *source universe* as well as in weighting, and a production-versus-arm difference cannot be attributed to novelty alone.
Two consequences, and both belong here rather than in §9. Comparisons *within* the grid — cell against cell, one aggregation scheme against another — are clean, because those arms share the same source universe and differ only in the score; that is where the paper's findings live (§7). Comparisons against production are confounded and should be read as context rather than as measurement. Rebuilding the arms on source-filtered counts is what would make the original comparison valid, and it is cheap; it is listed in §10.

6.4 Sample period

[plan] 518 trading days, 2024-07-01 to 2026-07-20. Benchmark: published STOXX 600, which stops 18 trading days before the price panel ends on 2026-08-28; beta and alpha require a market return, so those days are excluded from every figure.
Main text, not a footnote. Over the 18 excluded days production returned +50.1% annualised against a 45-cell mean of +6.6%. Including them moves production's Sharpe from 0.893 to 0.982 and the cell mean from 0.878 to 0.867 — *reversing the Sharpe comparison*. The beta result is unaffected. It is the clearest single illustration of how little 518 days can settle, and it is why §5 demotes Sharpe.

7 Results

[plan] Report, do not editorialise — interpretation is §8. **Organise by route**, per §3, so the section answers the paper's question rather than touring a notebook. All figures need re-export. **[TODO: export figures to `fig/` as PDF; several notebook cells exported empty HTML and must be re-run with outputs]**

7.1 Baseline decomposition

[plan] Regressing daily returns on the STOXX 600: $R^2 = 77.3\%$, intercept -0.26% p.a. with $t = -0.05$, Sharpe of the beta-hedged stream -0.038. Neutral and brief: over this window the production leg's realised return is predominantly market exposure, which is why the comparison below is run on beta and alpha rather than on headline return. Do not dwell.

7.2 Weighting

[plan] Weight mode is where the pilot has the most evidence, so lead with it.
The headline, and it is negative. On the two windows scanned, no cell of the grid clears production's information coefficient at any horizon — 0 of 18 on the raw score and 0 of 18 on the corrected score. Report the full table, because the level matters as much as the comparison:

IC at horizon	h=1	h=5	h=10	h=21	h=63
Production	0.0000	−0.0044	−0.0051	−0.0131	−0.0264
Raw, mean of cells	−0.0022	−0.0122	−0.0136	−0.0248	−0.0376
Norm, mean of cells	−0.0018	−0.0107	−0.0116	−0.0222	−0.0343
Raw, best cell	−0.0009	−0.0076	−0.0077	−0.0169	−0.0250
Norm, best cell	−0.0009	−0.0077	−0.0077	−0.0168	−0.0247

Two things to say about this and not gloss over. First, every arm including production has an information coefficient at or below zero on this window, and it grows more negative with horizon — so the comparison is between a signal with no measured cross-sectional edge and variants of it with slightly less. Second, novelty weighting moves every cell in the *same* direction, away from production, at every horizon. On the primary endpoint of §5.1, the weight route did not enhance the signal on this sample.
But the Sharpe picture disagrees, and the disagreement is the finding. Cell Sharpe declines steeply in n, from roughly 1.05 at $n = 1$ — above production's 0.893 — to 0.6–0.75 at $n = 100$; the best cell reaches 1.059. So small-n cells beat production on Sharpe while beating it on IC in zero cases. A Sharpe gain with no ranking gain has to come from the portfolio side rather than the signal side, and §7.4 names the candidate: these cells carry lower market beta. **That makes the beta attribution diagnostic the highest-priority open item in the paper** — if the Sharpe advantage is a beta de-tilt in a rising market, the weight route's apparent success is not about novelty at all. **[TODO: run the beta attribution notebook before this section is written up]**

7.3 Filtering

[plan] Filter mode with θ swept, and the matched-keep-rate version of the same grid at matched keep-rate — each cell given its own threshold so all cells trade the same fraction of articles, which is what isolates lookback from filter strength (§3.1). Report both.

7.4 Market beta

[plan] 45 of 45 cells carry a lower beta than production; cell betas span 0.793–0.834 against production's 0.894. Caveat in the same paragraph: the 45 cells are 45 settings of one measurement and neighbouring settings share most of their neighbours, so this is a uniform directional effect with no counterexample in the grid — not a sign test, and no p-value should be attached to it.
Then the three candidate explanations, per H5: exposure (cash weight changes with the z distribution), size tilt (novelty and coverage intensity are mechanically correlated, and coverage correlates with size — Fang and Peress, 2009, Hillert et al., 2014), and selection. **[TODO: run the attribution diagnostic before the paper commits to the selection reading]**

7.5 Alpha and the neighbour count

[plan] On the published five-window grid: Spearman correlation between n and mean alpha $= -0.800$, $p = 0.0096$; mean alpha positive at $n \in \{1,3,5,7\}$ and negative at $n \in \{10,20,50,70,100\}$. This is the pilot's clearest rank result and the one the sample length can actually support (§5.1). Interpretation in §8.1.

7.6 Risk–return characteristics

[plan] The composite is the equally weighted average of the daily returns of the 12 cells in windows 3d/7d/14d at $n \in \{1,3,5,7\}$ — selected by the rank ordering of §7.5, not by these outcomes. Say so — it is the only thing keeping the composite from being an in-sample pick.
Composite against production: return $+12.89\%$ vs $+12.73\%$; volatility 13.27% vs 14.25%; Sharpe 0.971 vs 0.893; beta 0.822 vs 0.894. Sharpe is invariant to leverage, so position along a constant-Sharpe ray carries no information and only position between rays does.

7.7 Paired bootstrap

[plan] Moving-block bootstrap (block 10 trading days, 4,000 draws) of the paired daily difference between each arm's beta-hedged stream and production's. Every interval spans zero; no cell reaches $|t(\alpha)| > 2$. Point to §5.1 rather than restating the arithmetic here.

7.8 In-sample threshold optimisation

[plan] In-sample θ optimisation, one Optuna TPE study per arm through the same production book. θ^* is not comparable across arms — a longer window pulls neighbours closer so novelty shifts down and a fixed θ filters harder, while a larger n averages in weaker matches so novelty shifts up and the same θ filters more loosely. What is comparable is each arm's best achievable Sharpe. Label it an optimistic ceiling, not a tradeable setting, and say what it is for: bounding headroom before spending on the expensive candidates. And state the reading rule once, here, because this is where it is most needed: the grid is one family of comparisons rather than dozens of independent experiments, so the reported quantity is the grid-level result and the maximum cell is descriptive only.

8 Discussion

[plan] Separate what the sample supports from what it suggests. The section a future reader will read first, so it must stand alone. Two pages, organised by route so it answers the paper's question directly.

8.1 Findings by aggregation scheme

[plan] Two results survive the sample, and then what each route can be said to have shown. Structure it that way, because the routes are the question the paper asked.
The two results that survive. First, the beta reduction (§7.4): uniform across the grid and insensitive to the excluded-window disclosure — though which of the three candidate explanations owns it stays unresolved until the attribution diagnostic runs. Second, the monotone ordering of alpha in the neighbour count (§7.5): a rank result over a swept parameter, far more robust at this sample length than any level comparison.
What the ordering means, in one paragraph and not more. Small neighbour counts ask *is this a copy of something already published?* Large ones ask *how crowded is recent coverage of this company?* Those are different questions and the parameter silently chooses between them, so the ordering is a finding about what novelty is for this purpose: redundancy is a property of the nearest one or few prior items, and averaging over the twentieth-nearest estimates something else. Two consequences — work at the strict-duplicate end, and prefer a formulation that fixes an *effective* neighbour count per article over a global one, since a company with three articles a day and one with three thousand are not asking the same question at the same setting. That is the only roadmap priority the pilot speaks to directly (§10.4).
Filter. What the swept threshold and the matched-keep-rate grid show, and the honest note that a positive result here replicates RavenPack Data Science Team (2014) and Ke et al. (2019) rather than extending them.
Weight. Where the pilot has the most evidence and the literature the least — and where the negative information-coefficient result of §7.2 therefore lands hardest. Read it against the Sharpe–IC disagreement rather than on its own.
Select. Untested, and the best-powered endpoint available (§5.1). This is the argument for prioritising the originator stage over further parameter sweeping.

8.2 Divergence from the first-story-detection literature

[plan] Vuurens and de Vries (2016) found 3-nearest-neighbour beats single-nearest for first-story detection; our alpha ordering points the other way. Different objectives — detection accuracy against event labels versus cross-sectional return prediction — and the divergence is informative: the k that best detects a first story need not be the k that best weights a sentiment signal. Do not paper over it.

8.3 Claims the sample cannot support

[plan] Any claim about alpha or Sharpe — point to §5.1 and §6.4, and note that the whole window is a markedly better-than-average stretch of market for this strategy.
What the embedding treats as similar, which is the construction's largest unexamined assumption and was moved here from §2.3. The vectors are cosine-comparable because they were trained to be, on a similarity notion that is not "reports the same event" (§2.3), and nothing in this paper establishes how far the two diverge on news. Name the two consequences. Every novelty figure reported is conditional on that correspondence holding, and the anisotropy this paper leaves uncorrected (§9) moves the whole similarity distribution by an amount fixed by the model. The intrinsic evaluation of §10 is the test, and it needs no return data.
Untested: the select route entirely; rank calibration; multilingual behaviour; the deferred candidates of Appendix D; the cost–signal frontier; and everything in Appendix E.

9 Limitations

[plan] Placed before §10 deliberately, so that the roadmap reads as a response to these rather than as an independent wish list. **[reuse: paper3 §8]** verbatim: wire syndication (separate strata; press releases identified and excluded at ingestion, since being first to restate a public disclosure measures wire latency rather than origination), embedding anisotropy, look-ahead bias (retrospective entity links, silently revised articles, calibration distributions estimated over the full window), coverage bias (an outlet can only be first among outlets that are crawled — the claim is "first within the observed universe"; quantify by re-running attribution on random subsets of $\mathscr{O}$), and model non-stationarity.
Add six, specific to this paper:

- **Sample length.** 518 days cannot resolve a Sharpe difference of 0.08; SE $\approx \pm 0.70$. §5.1.
- **Regime.** A markedly better-than-average stretch of market, and the 18 excluded days flip the Sharpe comparison (§6.4).
- **Cell dependence.** The 45 cells are 45 settings of one measurement, not 45 independent trials; neighbouring settings share most of their neighbours. Counts such as "45 of 45" are uniform directional effects, not sign tests.
- **In-sample θ.** §7.8 is a ceiling by construction.
- **Chronological consistency.** The embedding model has read the news it scores, and being a hosted proprietary model it cannot be pinned to a version. Tolerable over two years; the primary hygiene risk of any longer backfill, and the argument for a chronologically consistent model (He et al., 2025) before extending the sample.
- **The notebooks do not share an evaluation window.** The beta and alpha decomposition is measured over the 518 trading days to 2026-07-20 that the benchmark file also covers, while the threshold and grid work extends to the end of the price panel on 2026-08-28, which only the arms reach. Every figure in §7 is internally consistent, but figures should not be quoted across notebooks without naming the window — the excluded stretch of §6.4 is large enough to reverse a comparison on its own. Appendix F records which window each notebook uses.
- **Company size is not observable in this corpus.** Coverage volume correlates with company size and with attention, so a novelty adjustment may interact with size. The interaction cannot be tested here. `IndexConstituent` exposes name, ISIN, RIC and identifier only, the price panel is close-only, so neither market capitalisation nor traded value is available, and market-cap rank exists only as a server-side universe filter. Article flow is used as the coverage proxy throughout and is not a size measure.
- **Near-zero baseline IC.** On this window production's own information coefficient is 0.0000 at $h{=}1$ and negative at longer horizons (§7.2). Comparisons against a baseline with no measured cross-sectional edge are weak tests in both directions, and the paper should not claim a novelty variant is worse than a signal that is itself not distinguishable from zero.
- **Score reproduction drift.** The rescan renormalises embeddings to exactly unit length, so recomputed raw scores differ from `novelty_scores_wv.parquet` by about 1.3×10^{-3} per article and panel sums by about 0.08%. Immaterial for rank results; state it anyway.

10 Further work

[plan] NOT FINALISED. Written out, but the ordering and the priorities below are a proposal for review, not a settled plan. Everything here is ordered by dependency, and each item names the constraint that blocks the items after it.

The items below are ordered by dependency. Each names the constraint that blocks the ones after it.

10.1 Rebuilding the arms on source-filtered counts

The arms are built from the unfiltered article dump, while the production signal is filtered to outlets with $\tau_o \geq 2.5$ (§6). Every production-relative number in §7 therefore compares two different source universes, and each is provisional until the arms are rebuilt on production's own universe. That rebuild needs one re-run and no new data, which is why it comes first.

10.2 Extending the sample

Novelty scores begin in July 2024 and cover 518 trading days, against the 2015–2025 window and roughly 22,000 companies and 241,000 outlets of Khan (2026). Little in §7 or Appendix E becomes decisive at the present length, so the backfill is the precondition for most of what follows.

Three sub-tasks stand in a fixed order. The embedding model has to be pinned or replaced before any vector is written, because a backfill run while the hosted model drifts yields a history that cannot be compared against itself, and He et al. (2025) is the chronologically consistent alternative. Whatever is chosen, do not reach for a translation-pair objective such as Feng et al. (2022): it is trained on sentences paired with their translations, so it places a translated republication close to its source while leaving two articles written separately about one event far apart — which is the harder and more valuable case (§1.1). Storing the model version alongside every vector, and refusing to mix versions inside one reference set, is what makes that guarantee checkable afterwards. The backfill itself must then be strictly point-in-time, closing the three leakage channels of §9. Only then does the index choice arise, and Malkov and Yashunin (2020) suits an append-only stream under a sliding window because HNSW admits incremental insertion. Expiring a window is its documented weak point and needs a decision recorded before the backfill runs, since the alternative is periodic full rebuilds at a cost that scales with the history.

10.3 Identifying originators

Originator identification is placed above any further parameter sweeping. Attribution accuracy against the self-declared labels is a classification problem with tens of thousands of instances and no dependence on return data (§5.1), so it reaches a decisive answer on the one route the pilot leaves untested without waiting for the backfill. It requires the criterion of Appendix C and the attribution-phrase labels, and nothing else.

10.4 Building the remaining candidates

Priority here follows the pilot's own finding (§8.1) and not build cost. N3 with a perplexity-targeted temperature comes first (Appendix D), as the direct instrument for the narrow neigh-

bourhood in which the pilot locates the signal and as a principled replacement for a globally fixed n. N4 needs no build at all (Appendix B). N5, N6 and N7 follow, the last of them salience-weighted. The lead-weighted ablation of §2.2 and rank calibration per entity and language apply across every candidate.

10.5 The comparative protocol

The protocol benchmarks all three routes against production and against cheaper alternatives, holding everything but the map from $A_{c,o,t}$ to $S_{c,o,t}$ fixed. Universe, evaluation dates, forward-return construction and the cross-sectional rank information coefficient stay constant across arms, so that a difference in an endpoint is attributable to the aggregation and to nothing else.

Three baselines and three treatments are required. B0 is the production signal. B1 is deduplication alone, by locality-sensitive hashing and no embedding, which isolates the share of any gain attributable to verbatim syndication. B2 is the lexical novelty baselines, Zhang et al. (2002)'s TF–IDF maximum cosine and Karkali et al. (2013)'s stream IDF statistic, which establish whether the embedding earns its cost. T1, T2 and T3 are the filter, weight and select routes of §3, each crossed with each candidate. Reported endpoints are those of §5.1, together with the surviving quality-panel size at $\tau_o \geq 2.5$ and the company coverage, because a variant that improves an endpoint while halving the investable universe has not improved the signal.

One diagnostic is cheap and worth reporting alongside the endpoints, because §1.2 predicts it and nothing here has measured it. Estimate the originating share of a company-day as a function of that day's article count. The prediction is that it falls as coverage broadens, which if it holds means a small source set suffices on widely covered events and breadth earns its cost only on thinly covered ones. That is also the sharpest available statement of what the select route buys, and it needs no return data.

Two tests belong inside the protocol and not after it. The orthogonality test (§5.2) estimates the originating rate and measures its association with $\max_h \tau_o(h)$, and it is the cheapest available falsification of the whole project, so it runs before anything expensive is built. The horse race of Didisheim et al. (2026) then places novelty against the existing quality panel as two orthogonalisation axes on one dependent variable. A result on that axis will be demanded of this work whether or not the paper runs it.

10.6 Intrinsic evaluation

Every endpoint above is extrinsic, so a candidate that ranks articles badly and happens to help a portfolio cannot be distinguished from one that ranks them well. Two label sources already exist in the corpus. Attribution phrases give self-declared originator labels, and syndication clusters surviving near-duplicate removal give a negative set by construction. TAP-DLND 1.0 (Ghosal et al., 2018) then serves as an external sanity check on ranking behaviour, and SemEval-2022 Task 8 (Chen et al., 2022) as the instrument for the question of which similarity dimension the cosine actually tracks.

10.7 The cost–signal frontier

The information coefficient as a function of the selection ratio r is the quantity a production decision would need, and the pilot already sits at the cheap end of it, so the frontier is built downward from ϕ_{full}. It is placed late deliberately. Optimising the cost of a signal whose incremental content is not yet established would fix a cost target to an unproven benefit.

10.8 Design changes indicated by the literature

Three changes are indicated by work published since the pilot was specified, in priority order.

1. Multilingual Matryoshka embeddings (Hanley and Durumeric, 2025, Kusupati et al., 2022), in which similarity granularity is read off the prefix length. A fixed cosine threshold cannot separate the same story from the same event from the same topic, and prefix length gives that distinction directly. The method is state of the art on SemEval-2022 Task 8, operates in this paper's domain, and composes with the existing cascade. If one new technique is adopted, this is the one.

2. Salience weighting of sentence units (Ai et al., 2024), applied before the sentence-coverage candidate ships and not after.

3. A conventionality and novelty pair in place of a single scalar (Uzzi et al., 2013). The neighbour distribution is already computed and its concentration supplies most of the conventionality half, so the cost is small.

A Notation

Every symbol below is introduced in the text, in §2.1 unless a later section is named. The table is a lookup, and no argument in the paper depends on reading it.

Letters were chosen to name what they hold. $\mathscr{D}$ is a corpus of documents and d one of them, $C(d)$ the companies of d and c one of them, and $\mathscr{R}$ the reference set. ν is a Greek n for novelty, θ a threshold, τ_o a Greek t for an outlet's t-statistic, s a similarity, A a set of articles and S the signal built from it. ϕ and ψ are the two maps applied in that order, text selection then embedding. A tilde marks a calibrated quantity, a bar a mean, a hat an estimate, and a parenthesised subscript an order statistic.

A semicolon separates what an object depends on by definition from what a choice fixes. The reference set of d is $\mathscr{R}_c(d)$ whatever rule produces it, and $\mathscr{R}_c(d;w)$ once the lookback duration is fixed, so a symbol written without its parameters is the general object and one written with them is the instance used here. A parenthesised superscript does the same work for the neighbour count, as in $\bar{s}_c^{(n)}(d)$.

Four conventions hold throughout. The neighbour count is n, and k is never used as an index. Every threshold is written θ with a subscript naming what it cuts. A tilde marks a calibrated quantity, as in $\tilde{\nu}_c(d)$ (§2.3). And every quantity depending on the company carries it as a subscript, dropped wherever context fixes the company, so that $\nu(d)$ abbreviates $\nu_c(d)$ and $s_{(j)}(d)$ abbreviates $s_{c,(j)}(d)$.

B Equivalence of the centroid and top-k mean scores

[plan] Here only because the collapse of two candidates into one is used in §2.5 and §10, and the claim should be checkable. Nothing in the paper's argument depends on reading it.

All embeddings are taken to be L^2-normalised, so $x_d \in \mathbb{S}^{m-1}$ and $s(d,d') = x_d^\top x_{d'}$.[1] Let $x_{(1)},\dots,x_{(n)}$ be the embeddings of the n nearest prior neighbours of d in $\mathscr{R}_c(d;w)$, ordered by

[1] Exactly in the derivation, approximately in the data. The stored vectors are 256-dimensional, dimension-reduced by the provider from the model's native 1536, and are unit-norm only to about 2.5×10^{-4}. Both facts

Symbol	Meaning
Articles	
$\mathscr{D}$	the corpus of articles
d	an article
$t(d)$, $o(d)$, $C(d)$	its timestamp, its outlet, the companies it mentions
$\mathscr{D}_t$	the corpus as it stood at time t
c	a company in $C(d)$, against whose coverage d is scored; carried as a subscript on every company-dependent quantity
Representation and similarity	
ϕ	representation map, in the variants ϕ_{head}, ϕ_{comp} and ϕ_{full} (§2.2)
$x_d \in \mathbb{S}^{m-1}$	the unit-norm embedding of $\phi(d)$, in m dimensions
$s(d,d')$	cosine similarity, equal to $x_d^\top x_{d'}$ under unit norm
Reference set and novelty	
w	lookback duration
$\mathscr{R}_c(d;w)$	the reference set of d with respect to company c
n	neighbour count, at most $n_c(d) = \lvert\mathscr{R}_c(d;w)\rvert$
$s_{c,(j)}(d)$	the j-th largest similarity between d and its reference set
$\bar{s}_c^{(n)}(d)$	the mean of the n largest such similarities
$\nu_c(d)$, $\tilde{\nu}_c(d)$	raw and calibrated novelty of d with respect to c
$\widehat{G}_c$	the empirical distribution against which $\nu_c(d)$ is calibrated (§2.3)
θ_ν	novelty cutoff, in filter mode
Signal and evaluation	
h	forward horizon in trading days
tone(d)	the tonality assigned to article d
$A_{c,o,t}$	the articles about c published by outlet o on day t
$S_{c,o,t}$	the signal aggregated from $A_{c,o,t}$
z	cross-sectional robust z-score
τ_o	the outlet's t-statistic in the quality panel

Table 2: Symbols used in the main text, in roughly the order §2.1 introduces them. Quantities confined to a single appendix are defined where they appear.

decreasing similarity, and write $s_{(j)}(d) = x_d^\top x_{(j)}$. The top-$k$ mean candidate is

$$\nu_n(d) \;=\; 1 - \frac{1}{n}\sum_{j=1}^{n} s_{(j)}(d).$$

By linearity of the inner product, the mean of the similarities is the similarity to the mean:

$$\frac{1}{n}\sum_{j=1}^{n} s_{(j)}(d) \;=\; \frac{1}{n}\sum_{j=1}^{n} x_d^\top x_{(j)} \;=\; x_d^\top\Big(\frac{1}{n}\sum_{j=1}^{n} x_{(j)}\Big) \;=\; x_d^\top \bar{x}_d^{(n)}, \tag{2}$$

where $\bar{x}_d^{(n)}$ is the centroid of the n nearest neighbours. Averaging similarities and averaging neighbour vectors are therefore the same operation performed in a different order: the top-k mean is already a centroid score, taken over a neighbourhood rather than over the whole reference set.

are immaterial to the identity below — it needs only linearity of the inner product — but the polar decomposition and the $\bar{r}^{(1)} = 1$ limit hold to that tolerance rather than exactly, and any implementation checking them should compare against a tolerance rather than against zero.

Writing the centroid in polar form $\bar{x}_d^{(n)} = \bar{r}_d^{(n)} \hat{x}_d^{(n)}$ with $\bar{r}_d^{(n)} = \|\bar{x}_d^{(n)}\|_2 \in [0,1]$ and $\hat{x}_d^{(n)} \in \mathbb{S}^{m-1}$,

$$\nu_n(d) \;=\; 1 - \bar{r}_d^{(n)} \cos\angle\big(x_d, \hat{x}_d^{(n)}\big), \qquad \big(\bar{r}_d^{(n)}\big)^2 = \frac{1}{n} + \frac{n-1}{n}\,\bar{s}_d^{(n)}, \tag{3}$$

with $\bar{s}_d^{(n)}$ the mean pairwise similarity among the n neighbours, so $\bar{r}_d^{(n)}$ increases with the internal coherence of the neighbourhood and does not depend on d except through which neighbours it selects.

Two limits follow. At $n = 1$ the centroid is a single unit vector, so $\bar{r}_d^{(1)} = 1$ and $\nu_1(d) = 1 - s_{(1)}(d)$: the nearest-neighbour candidate, with no factor. At $n = n_d = |\mathscr{R}_c(d;w)|$ the neighbourhood is the entire reference set, $\bar{x}_d^{(n_d)} = \bar{x}_d$, and

$$\nu_{n_d}(d) \;=\; 1 - x_d^\top \bar{x}_d,$$

which is the naive centroid score. The centroid candidate scores against the mean *direction* instead, $\nu_4(d) = 1 - x_d^\top \hat{x}_d$, and the two similarity terms differ by exactly one multiplicative factor:

$$1 - \nu_4(d) \;=\; \frac{1 - \nu_{n_d}(d)}{\bar{r}_d}. \tag{4}$$

[plan] The conclusion to state, and its precise limits. By (2)–(4) the centroid candidate is the top-k mean evaluated at the widest neighbourhood, up to the factor $\bar{r}_d$. It is therefore not an independent measure and needs no separate implementation: it differs from the implemented candidate in two parameters — neighbourhood width and whether the factor is divided out — and in nothing else.
Be precise about "up to a factor". Since $\bar{r}_d$ depends only on the reference set, the two scores induce the *same* ordering among articles scored against the same reference set, and different orderings across reference sets. So they are interchangeable within a company-window and not interchangeable cross-sectionally — which is the only sense in which the distinction could have mattered.
Note also that for thinly covered entities the widest setting is reached at modest n, since the top-n set exhausts $\mathscr{R}_c(d;w)$: for those entities the implemented candidate *is* the centroid candidate well before the neighbour count reaches its largest values. Keep Karkali et al. (2013) cited — the document-to-summary formulation is the prior occurrence of this corner.

C Proposed originator criterion

[plan] Outlook, and prose only. This is the design for the select route of §3.3. Nothing here is implemented, so it introduces no notation — the formal criterion already exists in `novelty_scoring_paper3.tex` §§5.8–5.10 and can be lifted verbatim when the work is scheduled. Two pages at most, written so that someone picking the project up knows what to build and what will go wrong.

The criterion. Two conditions rather than one. An article must be novel with respect to the coverage that preceded it, and it must be echoed by the coverage that follows — meaning later articles about the same entity are closer to it than to anything published before it. The signature to detect is a spike in novelty followed by its collapse: one article occupies a previously unoccupied region of the embedding space, and subsequent coverage converges into it. Both a novelty threshold and a minimum echo count are required, and both are free parameters.

Why novelty alone is insufficient. The set of articles most dissimilar from their own past is populated by machine-translation artefacts, off-topic pieces that mention an entity in passing, and low-quality reproduction. All score as novel; none is a first report. Subsequent coverage is the discriminator, which is also why the criterion is irreducibly retrospective.

What it yields at outlet level. Three quantities, and the important point is that they are not interchangeable: how often an outlet originates, how far ahead of the echo it publishes, and whether the stories it originates are followed by abnormal returns. An outlet can originate frequently, with a long lead, on stories that never move a price. Outlet-level estimates need shrinking toward the universe mean in proportion to how much that outlet publishes, and should be required to hold up out of sample.

Why it cannot be applied live. The criterion needs the article's future, so it classifies retrospectively. Live use means treating an outlet's historical originating rate as a prior on its incoming articles — which is the operational reason outlet-level scoring exists at all, rather than article-level.

Evaluation without manual annotation. Two sources of ground truth. Explicit attribution phrases — "as first reported by X" — are conventional in financial journalism, extractable by pattern matching, and give self-declared originator labels; they validate the echo criterion directly, which a returns-based evaluation does not. And after near-duplicate removal, verbatim republications are known-redundant by construction, so any of them scored as originating is a false positive. Together these make the select route the best-powered endpoint available (§5.1), since it becomes a classification problem with many thousands of instances and needs no return data at all.

Three things that will go wrong.

- **Common-source overlap** (Nicholls, 2019). Two outlets independently rewriting the same filing or wire flash produce a false originating edge, and the criterion cannot separate "B copied A" from "A and B both copied S" when S is outside the corpus. Either model the unobserved source (Gomez-Rodriguez et al., 2010) or state it as a limitation and mean it.
- **Single-parent assignment** (Kuntur et al., 2026). Measured reuse is one-to-one in only 4.4% of cases against 54.9% many-to-many, so assigning each article a single nearest predecessor models the rare case. Report how sensitive the originating rate is to allowing several.
- **Global averaging** (Stern et al., 2020). Intermedia influence is strongly topic-dependent — the same outlet leads on some subjects and follows on others — so a single global originating rate per outlet averages away the structure. Report per entity or per sector.

And one thing that is not a finding. Wire services will occupy the top of any originating ranking by construction, because originating and distributing news is their function. Wires and non-wire outlets need scoring in separate strata, and press-release restatement needs excluding at ingestion, since being first to restate a public disclosure measures wire latency rather than information origination (§9).

D Deferred candidate functions

[plan] **[reuse: paper3 §5.4, the N5, N6 and N7 paragraphs]** — definitions, failure modes and constraints only. **Drop** the cost derivations until one is scheduled; paper3 §5.6 has them if they are ever wanted.
Keep what is load-bearing for the roadmap:

- **N3, soft aggregation.** The one-parameter family interpolating N1 and N2 by softmax temperature, with the perplexity-targeted variant that fixes an *effective* neighbour count per article rather than a global n. It is deferred but it is not idle: §8.1 makes it the highest priority of the deferred set, because the pilot says the signal lives in the narrow-neighbourhood interval and a global n is a crude way to sit there. No canonical citation located — present it as an engineering choice, with the temperature limits recovering N1 and N2, and cite Wu et al. (2008) for why interpolating across the redundancy spectrum is motivated at all.
- **N5, subspace residual.** The recombination reading is the reason to keep it — Fedyk and Hodson (2023) show recombinations move prices more than reprints, and a recombination is an article far from every individual neighbour yet well inside the span of several. Note the prior art without softening it (Hoffmann, 2007; Mesarcik et al., 2022), and that it is the candidate most exposed to rogue dimensions (Timkey and van Schijndel, 2021), so whiten or standardise before projecting.
- **N6, time-decayed kernel.** Redundancy against an article published an hour ago is not equivalent to redundancy against one published twelve days ago. No canonical citation located; note the $O(1)$ EMA form.
- **N7, asymmetric sentence coverage.** Keep the asymmetry argument and both caveats from paper3. Add that asymmetry is not ours (Broder, 1997; Tsai et al., 2010) and that salience weighting is required before it ships (Ai et al., 2024).

Close with why all three are deferred: none is reachable before the backfill, and §8.1 says the next marginal unit of work belongs at the strict-duplicate end of the family — which is N3's interval, not N5's, N6's or N7's.

E The reporting-lag hypothesis

[plan] Wholly deferred, and moved here from the main text. The reason is worth stating rather than hiding: this is a claim about *horizons*, and the paper does not yet measure the horizon profile of its own signal — the IC-by-horizon tables of §7.2 are the missing piece. Proposing a mechanism for a horizon phenomenon before measuring horizons on our own data is the wrong order. Testing it also requires the originator stage of Appendix C, which does not exist either. Promote it to the main text when both exist and not before.
One sentence belongs in §8 so a reader who never reaches the appendix knows the idea exists: an outlet that reports with a systematic lag records its sentiment observation time-shifted by that lag, so its apparent optimal forecast horizon may in part measure its reporting delay rather than the diffusion speed of the information it carries — an alternative reading of the horizon-specificity reported by Khan (2026).

[plan] **[reuse: paper3 §4]** verbatim for the mechanism itself.

Let outlet o report a story with systematic lag Δ_o relative to the originating outlet. If the

underlying information is incorporated into prices over a horizon h^* measured from the originating report, then o attains its maximum information coefficient at $h^* + \Delta_o$ measured from its own publication date. Its apparent optimal horizon is then in part a measurement of Δ_o, not of the diffusion speed of the information it carries.

[plan] Then the three testable implications from paper3: (i) $\arg\max_h t_o(h)$ positively associated with mean reporting lag; (ii) the asymmetric decay reported by Khan (2026) is the expected signature of an outlet panel containing substantial time-shifted echo; (iii) the growth of the quality panel from roughly 8,500 outlets at $h = 1$ to 10,800 at $h = 63$ follows from echoing being more common than originating.

Evidence on publication-timestamp error

[plan] Bradley et al. (2014) is the anchor, quoted with its numbers: the *same* analyst recommendations give 30-minute returns of $+1.83\%/-2.10\%$ on newswire timestamps and $-0.07\%/-0.09\%$ on I/B/E/S timestamps — a pure timestamp error destroys the entire signal. Then Twedt (2016) (wire dissemination changes measured price-discovery speed), Hendershott et al. (2015) (institutional order flow predicts the occurrence and sentiment of Reuters news, so discovery precedes the timestamp), and Rogers et al. (2017) (even regulated EDGAR dissemination has a mean 40s lag, s.d. 24s, with large outliers).

Competing explanations

[plan] Not merely mention — rule out. Heston and Sinha (2017): temporal aggregation alone extends horizon, so the lag test must beat a naively time-aggregated signal. Fedyk (2021): gradual diffusion versus differences of opinion, separated with click-level data. Fedyk (2024): front-page placement produces large reactions for content that is algorithmically indistinguishable, so outlet-level results need a prominence control.

F Reproducibility

[plan] Make the work re-runnable by someone with no memory of it.
Notebooks: `novelty_alpha_decomposition.py` (beta/alpha decomposition, paired bootstrap, risk–return plane), `novelty_subset_ceiling.py` (IC by horizon, backtest Sharpe, in-sample θ study, per-company axis mechanics, grid surface), `novelty_dispersion_check.py` and `novelty_ceiling_normalised.py` (a score-construction correction, measured and found neutral on every outcome; not discussed in the body of this note, see PARKED-DISPERSION in the source). All marimo 0.24.0.
Entry points: `nk.build_arm`, `nk.PROD_PARAMS`, `build_sentiment`, `load_prices`, `load_counts`, `load_benchmark`, `Portfolio.truncate`, `Portfolio.stats`, `sentiment_ratio`, `_robust_signal`.
Production parameters as of this note: span 19, universe top-200, entry $z > 1.15$, 10% per-name cap, 10-day rebalance, 20% vol target, clip ± 4.2, outlet t-stat ≥ 2.5. Score files built with `max_cand=0`; grid 5 windows $\times$ 9 values of n. Daily return series per cell cached to parquet; figures read from cache. Dates: article scores from 2024-07-01; price panel to 2026-08-28; benchmark to 2026-07-20.
Exporting the notebooks. Several result cells have exported as empty HTML, leaving shared artefacts prose-only while the markdown survived. The export itself runs the notebook — there is no `--include-outputs` flag for `export html`, that option exists only on `export ipynb` and `export pdf` — so empty outputs mean either a cell errored during export or the output is a custom element the static bundle fills in only when a browser opens the file. Use

```
uv run marimo export html <notebook>.py -o <out>.html
```

run from the project root, so that the export resolves the project environment in which the notebook normally executes. A non-zero exit means some cells errored, and the HTML is still written with those errors embedded, so check the exit status rather than the file. **Do not add** `--sandbox`: it builds an isolated environment from the notebook's declared dependencies, which cannot resolve the internal imports these notebooks rely on, so every cell that touches them will error and export empty — the failure mode it is meant to rule out. If tables and charts are still blank when the file is opened, pre-render it with the Playwright recipe in marimo's static-HTML guide, or use `marimo export html-wasm <notebook>.py -o <dir> --mode run --execute` and serve the directory over HTTP — it will not render from a `file://` URL. Note that `mo.ui` element state is not preserved by HTML export in any case, so interactive controls export inert.

References

Lin Ai, Ziwei Gong, Harshsaiprasad Deshpande, Alexander Johnson, Emmy Phung, Ahmad Emami, and Julia Hirschberg. NovAScore: A new automated metric for evaluating document level novelty. Technical Report 2409.09249, arXiv, 2024. JPMorgan Chase ML Center of Excellence and Columbia University. Atomic-unit asymmetric coverage with salience weighting – the closest existing implementation of the idea behind N7.

James Allan, Ron Papka, and Victor Lavrenko. On-line new event detection and tracking. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '98)*, pages 37–45, 1998.

Kathleen Beckers, Andrea Masini, Julie Sevenans, Marlies van der Burg, Julie De Smedt, Hilde Van den Bulck, and Stefaan Walgrave. Are newspapers' news stories becoming more alike? media content diversity in belgium, 1983–2013. *Journalism*, 20(12):1665–1683, 2019. doi:

10.1177/1464884917706860. COUNTER-EVIDENCE, and cite it as such. Nine Flemish papers, 20,600 articles, one week each in 1983/1993/2003/2013. Mean pairwise story overlap rho = .092 (1983), .062, .062, .078 (2013): most alike in 1983, so the declining-diversity hypothesis is REJECTED. Ownership predicts similarity (B = .122, p ¡ .001); strongest pair rho = .415. Regional sections were EXCLUDED from the sample and Belga is never mentioned, which bounds what it refutes.

Elizabeth Blankespoor, Ed deHaan, and Ivan Marinovic. Disclosure processing costs, investors' information choice, and equity market outcomes: A review. *Journal of Accounting and Economics*, 69(2–3), 2020.

Jelle Boumans, Damian Trilling, Rens Vliegenthart, and Hajo Boomgaarden. The agency makes the (online) news world go round: The impact of news agency content on print and online news. *International Journal of Communication*, 12:1768–1789, 2018. URL https://ijoc.org/index.php/ijoc/article/view/7109. Netherlands, one year of ANP output (n = 119,452 items) traced into national print and online outlets (n = 247,161 articles) by automated content analysis. Up to 75% of ONLINE articles attributable to the agency, much carried verbatim; print is lower. NATIONAL outlets only – says nothing directly about regional or local outlets. IJoC assigns no DOIs; do not invent one.

Daniel Bradley, Jonathan Clarke, Suzanne S. Lee, and Chayawat Ornthanalai. Are analysts' recommendations informative? intraday evidence on the impact of time stamp delays. *The Journal of Finance*, 69(2):645–673, 2014.

Andrei Z. Broder. On the resemblance and containment of documents. In *Proceedings of the Compression and Complexity of Sequences (SEQUENCES '97)*, pages 21–29. IEEE, 1997. Broder's "containment" is the lexical asymmetric measure – the 1997 counterexample to any claim that N7's asymmetry is new.

Brian J. Bushee, John E. Core, Wayne R. Guay, and Sophia Hamm. The role of the business press as an information intermediary. *Journal of Accounting Research*, 48(1):1–19, 2010.

Xi Chen, Ali Zeynali, Chico Camargo, Fabian Flöck, Devin Gaffney, Przemyslaw Grabowicz, Scott A. Hale, David Jurgens, and Mattia Samory. SemEval-2022 task 8: Multilingual news article similarity. In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*. Association for Computational Linguistics, 2022. Seven annotated similarity dimensions: geography, narrative, time period, tone, entities, style, overall.

Zhi Da, Joseph Engelberg, and Pengjie Gao. In search of attention. *The Journal of Finance*, 66 (5):1461–1499, 2011.

Antoine Didisheim, Bryan T. Kelly, Mo Pourmohammadi, and Hanqing Tian. The inefficient pricing of news. NBER Working Paper 35093, National Bureau of Economic Research, 2026. Also SSRN 6610736.

Michael S. Drake, Nicholas M. Guest, and Brady J. Twedt. The media and mispricing: The role of the business press in the pricing of accounting information. *The Accounting Review*, 89(5):1673–1701, 2014.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of EMNLP-IJCNLP 2019*, pages 55–65, 2019.

Lily Fang and Joel Peress. Media coverage and the cross-section of stock returns. *The Journal of Finance*, 64(5):2023–2052, 2009.

Anastassia Fedyk. Disagreement after news: Gradual information diffusion or differences of opinion? *Review of Asset Pricing Studies*, 11(3):465–501, 2021.

Anastassia Fedyk. Front-page news: The effect of news positioning on financial markets. *The Journal of Finance*, 79(1):5–33, 2024.

Anastassia Fedyk and James Hodson. When can the market identify old news? *Journal of Financial Economics*, 149(1):92–113, 2023.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL 2022), Volume 1: Long Papers*, pages 878–891, 2022. LaBSE is optimised for translation-pair retrieval, a narrower relation than "reports the same event". Do not treat as a neutral default.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations (ICLR 2019)*, 2019.

Tirthankar Ghosal, Amitra Salam, Swati Tiwari, Asif Ekbal, and Pushpak Bhattacharyya. TAP-DLND 1.0: A corpus for document level novelty detection. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. ELRA, 2018.

Paul Glasserman and Harry Mamaysky. Does unusual news forecast market stress? *Journal of Financial and Quantitative Analysis*, 54(5):1937–1974, 2019.

Manuel Gomez-Rodriguez, Jure Leskovec, and Andreas Krause. Inferring networks of diffusion and influence. In *Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '10)*, pages 1019–1028, 2010. NetInf. Extended version: ACM TKDD 5(4), 2012. The principled multi-parent alternative to our greedy single-parent echo rule.

Hans W. A. Hanley and Zakir Durumeric. Hierarchical level-wise news article clustering via multilingual matryoshka embeddings. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025), Volume 1: Long Papers*, 2025. ACL Anthology 2025.acl-long.124; arXiv:2506.00277. Story-similarity granularity read off embedding prefix length. SOTA on SemEval-2022 Task 8.

Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. Technical Report 2502.21206, arXiv, 2025. Also SSRN 5159615. Introduces ChronoBERT / ChronoGPT.

Terrence Hendershott, Dmitry Livdan, and Norman Schürhoff. Are institutions informed about news? *Journal of Financial Economics*, 117(2):249–287, 2015.

Steven L. Heston and Nitish Ranjan Sinha. News vs. sentiment: Predicting stock returns from news stories. *Financial Analysts Journal*, 73(3):67–83, 2017.

Alexander Hillert, Heiko Jacobs, and Sebastian Müller. Media makes momentum. *Review of Financial Studies*, 27(12):3467–3501, 2014.

Heiko Hoffmann. Kernel PCA for novelty detection. *Pattern Recognition*, 40(3):863–874, 2007. Reconstruction error w.r.t. a principal subspace IS the novelty measure. N5's measure, published 2007. Our claim must be narrowed accordingly.

Benjamin D. Horne and Sibel Adalı. An exploration of verbatim content republishing by news producers. In *NECO 2018 Workshop, co-located with ICWSM 2018*, 2018. arXiv:1805.05939.

Margarita Karkali, François Rousseau, Alexandros Ntoulas, and Michalis Vazirgiannis. Efficient online novelty detection in news streams. In *Web Information Systems Engineering (WISE 2013)*, volume 8180 of *Lecture Notes in Computer Science*, pages 57–71. Springer, 2013.

Zheng Tracy Ke, Bryan T. Kelly, and Dacheng Xiu. Predicting returns with text data. NBER Working Paper 26186, National Bureau of Economic Research, 2019. Also Yale ICF WP 2019-10 / Chicago Booth RP 20-37. WORKING PAPER – no peer-reviewed publication located as of Aug 2026.

M. D. Khan. Sentiment decay and source selection in global equity markets. White paper, Yukka AG, 2026.

Sara Kuntur, Nina Smirnova, Anna Wróblewska, Philipp Mayr, and Sara Razboršek Maček. Rewrite the news: Tracing editorial reuse across news agencies. In *SoCon-NLPSI 2026 Workshop @ LREC 2026*, 2026. arXiv:2603.29937. Reuse is 1:1 in only 4.4% of cases, many:many 54.9%; article leads are the most original section. Dataset and code at github.com/kunturs/lrec2026-rewrite-news.

Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi. Matryoshka representation learning. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.

Jure Leskovec, Lars Backstrom, and Jon Kleinberg. Meme-tracking and the dynamics of the news cycle. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '09)*, pages 497–506, 2009.

Yu. A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(4):824–836, 2020. Incremental insertion suits our append-only stream; window expiry (deletion) is the known weak point and needs a design decision.

R. Manmatha, T. Rath, and F. Feng. Modeling score distributions for combining the outputs of search engines. In *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '01)*, pages 267–275, 2001. doi: 10.1145/383952.384005. Fits a distribution to each source's raw scores and maps a raw score to its position in that distribution – the closest precedent for calibrating against a reference distribution rather than against a fixed cut-off.

Michael Mesarcik, Elena Ranguelova, Albert-Jan Boonstra, and Rob V. van Nieuwpoort. Improving novelty detection using the reconstructions of nearest neighbours. *Array*, 14:100182, 2022. doi: 10.1016/j.array.2022.100182. arXiv:2111.06150. The LOCAL neighbour version of subspace-residual novelty. Venue verified: Array (Elsevier), NOT Machine Learning with Applications and NOT Pattern Recognition Letters, both of which appear in secondary sources. Do not cite the duplicate SSRN record 10.2139/ssrn.4027210.

Takayuki Mizuno, Takaaki Ohnishi, and Tsutomu Watanabe. Novel and topical business news and their impact on stock market activity. *EPJ Data Science*, 2017. Earlier as arXiv:1507.06477 and CARF Working Paper F-366.

Mark Montague and Javed A. Aslam. Relevance score normalization for metasearch. In *Proceedings of the Tenth International Conference on Information and Knowledge Management (CIKM '01)*, pages 427–433, 2001. doi: 10.1145/502585.502657. The canonical statement that raw relevance scores are not on a comparable scale across sources and must be normalised before being combined or thresholded. Crossref also holds a duplicate front-matter DOI 10.1145/502653.502657 without pages – use the one above.

Tom Nicholls. Detecting textual reuse in news stories, at scale. *International Journal of Communication*, 13, 2019. Reuters Institute, University of Oxford. Four-way reuse taxonomy: newswire copy, PR material, source-to-source copying, common-source overlap.

Joel Peress. The media and the diffusion of information in financial markets: Evidence from newspaper strikes. *The Journal of Finance*, 69(5):2007–2043, 2014.

Victor Pickard. *Democracy Without Journalism? Confronting the Misinformation Society*. Oxford University Press, 2020. Ch. 3, "How Commercialism Degrades Journalism". DOI 10.1093/oso/9780190946753.001.0001.

RavenPack Data Science Team. Improved stock market returns from systematically trading infrequent news. Research note, RavenPack, October 2014. VENDOR RESEARCH, not peer reviewed. Event Similarity Gap ¿ 90 days applied to the Sentiment Strength Indicator raises long/short Sharpe from 1.03 to 1.43 on the Russell 2000, Jan 2007 – Aug 2014.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP 2019*, pages 3982–3992, 2019.

Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP 2020)*, pages 4512–4525, 2020.

Jonathan L. Rogers, Douglas J. Skinner, and Sarah L. C. Zechman. Run edgar run: Sec dissemination in a high-frequency world. *Journal of Accounting Research*, 55(2):459–, 2017.

Ian Soboroff and Donna Harman. Novelty detection: The trec experience. In *Proceedings of Human Language Technology Conference and Conference on Empirical Methods in Natural Language Processing (HLT/EMNLP 2005)*, 2005.

Samuel Stern, Giacomo Livan, and Robert E. Smith. A network perspective on intermedia agenda-setting. *Applied Network Science*, 5(31), 2020. Intermedia influence is strongly topic-dependent – so outlet scoop rate must be reported per entity or sector, not globally.

Paul C. Tetlock. All the news that's fit to reprint: Do investors react to stale information? *Review of Financial Studies*, 24(5):1481–1512, 2011.

William Timkey and Marten van Schijndel. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, pages 4527–4546, 2021. The single most dangerous pitfall for N1–N7; N5 is the most exposed.

Flora S. Tsai, Wenyin Tang, and Kap Luk Chan. Evaluation of novelty metrics for sentence-level novelty mining. *Information Sciences*, 180(12):2359–2374, 2010. Names and evaluates the symmetric vs asymmetric split for sentence-level novelty metrics – direct prior art on the asymmetry of N7.

Brady J. Twedt. Spreading the word: Price discovery and newswire dissemination of management earnings guidance. *The Accounting Review*, 91(1):317–346, 2016.

Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. Atypical combinations and scientific impact. *Science*, 342(6157):468–472, 2013. The impactful quantity was never novelty alone but a conventional base with an unusual intrusion – argues for a (conventionality, novelty) pair.

Daniel Vogler and Andrea Häuptli. Die bedeutung von nachrichtenagenturen für schweizer medien. Technical report, fög – Forschungsinstitut Öffentlichkeit und Gesellschaft, Universität Zürich, February 2018. URL https://www.foeg.uzh.ch/dam/jcr:accb3350-edef-4e00-a9e6-ed63f3416791/SDA_Analyse_final.pdf. NOT PEER-REVIEWED – research-institute report; say so if the number carries weight. Switzerland 2016–2017; 20 news sites and 26 newspapers, n = 41,377 items. Declared agency copy: regional press Basler Zeitung 41%, Berner Zeitung 37%, Südostschweiz 37%; regional sites nlz.ch 63%, tagblatt.ch 59%; from SDA specifically NZZ 13%, Le Temps 15%. Only 15% of declared agency items are local or regional in focus, 65% international. Counts DECLARED copy only, so every share is a floor – the report itself says the true figure is "weit grösser".

Jeroen B. P. Vuurens and Arjen P. de Vries. First story detection using multiple nearest neighbors. In *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '16)*, pages 845–848, 2016. 3-NN beats 1-NN for FSD – direct prior art for N2, and a result our pilot's alpha ordering points against.

Mengyu Wang and Tiejun Ma. MANA-Net: Mitigating aggregated sentiment homogenization with news weighting for enhanced market prediction. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*, pages 2379–2389, 2024. doi: 10.1145/3627673.3679653. arXiv:2409.05698. The ML-side statement of the aggregation problem: naive daily averaging homogenises sentiment, and learned per-article weights are the proposed fix.

William Webber, Alistair Moffat, and Justin Zobel. Score standardization for inter-collection comparison of retrieval systems. In *Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '08)*, pages 51–58, 2008. doi: 10.1145/1390334.1390346. CAVEAT: standardises effectiveness scores per topic, not document retrieval scores. A strong analogy for transfer across entities and periods, but do not describe it as document-score normalisation.

Wenqing Wu, Chengzhi Zhang, Tong Bao, and Yi Zhao. SC4ANM: Identifying optimal section combinations for automated novelty prediction in academic papers. *Expert Systems with Applications*, 2025. arXiv:2505.16330. Introduction + Results + Discussion beats whole text – motivates the lead-weighted ablation for N1–N7.

Xiao Wu, Alexander G. Hauptmann, and Chong-Wah Ngo. Measuring novelty and redundancy with multiple modalities in cross-lingual broadcast news. *Computer Vision and Image Understanding*, 110(3):418–431, 2008. No single novelty function dominates across degrees of redundancy – the precedent for reporting all seven candidates rather than picking one.

Yi Zhang, Jamie Callan, and Thomas Minka. Novelty and redundancy detection in adaptive filtering. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '02)*, pages 81–88, 2002.